CN Practical 7: Run Network Diagnostic Commands

Aim: To execute network diagnostic commands such as ipconfig, ping, tracert, netstat, route and arp.

Theory:

Network diagnostic commands are used to configure, inspect and troubleshoot computer networks. They help administrators identify IP configuration problems, test connectivity, examine network routes and view active connections. Windows provides several command-line utilities that can be executed using Command Prompt.

The ipconfig command displays IP configuration details. The ping command checks connectivity using ICMP echo requests. The tracert command identifies the intermediate routers between the source and destination. The netstat command displays network connections and listening ports. The route command displays or modifies the routing table, while arp displays the local Address Resolution Protocol cache.

Commands and their functions:

Command	Function
ipconfig	Displays IP configuration.
ipconfig /all	Displays detailed network information.
ping 127.0.0.1	Tests the local TCP/IP stack.
ping 8.8.8.8	Tests connectivity to a remote IP address.
tracert google.com	Displays the route to a destination.
netstat -an	Displays connections and listening ports.
route print	Displays the routing table.
arp -a	Displays the ARP cache.

Procedure:

1. Start the computer and ensure that the network adapter is enabled.
2. Open the Start menu and search for Command Prompt.
3. Open Command Prompt.
4. Enter ipconfig and observe the IPv4 address, subnet mask and gateway.
5. Enter ipconfig /all to view detailed network settings.
6. Execute ping 127.0.0.1 to test the local TCP/IP stack.
7. Execute ping 8.8.8.8 to check remote IP connectivity, if permitted.
8. Enter tracert google.com to view the route to the destination.
9. Enter netstat -an to display active connections and listening ports.
10. Enter route print to view the routing table.
11. Enter arp -a to display cached IP-to-MAC address mappings.
12. Record and analyse the output of each command.

Observation:

The diagnostic commands displayed different types of network information. The ipconfig command showed the IP configuration, while ping helped check connectivity. The tracert command displayed the route to a

destination, and netstat, route and arp provided details about connections, routes and address mappings. Some remote commands may fail if connectivity is restricted.

Conclusion:

Network diagnostic commands were executed successfully and their outputs were studied. These commands help identify configuration errors, connectivity problems and routing issues. They are useful tools for maintaining and troubleshooting computer networks.